

# Covering by Residue Classes

GREEN-085

L1 - TRACTABLE

OPEN

Number Theory

## ORIGINAL PROBLEM STATEMENT

Let  $N$  be large. For each prime  $p$  with  $N^{0.51} \leq p < 2N^{0.51}$ , pick a residue  $a(p) \in \mathbb{Z}/p\mathbb{Z}$ . Is  $\#\{n \in [N] : n \equiv a(p) \pmod{p} \text{ for some } p\} \gg N^{1-o(1)}$ ?

**Source and attribution.** Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.  
Archive: <https://www.unsolvedmath.com/problems?difficulty=1>  
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>